

SHENGZHI SUN, PhD

General Information

Dr. Shengzhi (“Darren”) Sun
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Education

2013-17	PhD	Epidemiology and Biostatistics	The University of Hong Kong
2010-13	MSc	Environmental Health	Peking University
2005-10	MBBS	Preventive Medicine	Wuhan University of Science & Technology

Academic Appointments

2020-	Research Scientist	Boston University School of Public Health
2019-	Research Associate	Brown University School of Public Health
2017-19	Postdoctoral Research Associate	Brown University School of Public Health
2016-17	Visiting Scholar	Harvard University School of Public Health
2013	Intern Analyst	The George Institute for Global Health

Teaching

2019	Teaching Associate	01 Introduction to Statistical Programming in R Brown University School of Public Health
2019	Teaching Associate	02 Introduction to Statistical Programming in R Brown University School of Public Health
2016	Teaching Assistant	Environmental Health Hazards and Interventions University of Hong Kong School of Public Health
2016	Teaching Assistant	Environmental Health Assessment University of Hong Kong School of Public Health

Research Grants

2019-2022	Evaluation of Municipal Heat Action Plans (HAPs) in the United States (PI: Wellenius) Wellcome Trust Role: Co-I
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Grant Support

2018-2022	Evidence to improve heat warning effectiveness in reducing morbidity and mortality (PI: Wellenius) NIH R01 ES029950 Role: Research Associate
2017-2019	Health Effects of Long-term Climatic Changes (PI: Wellenius) Seed Funds from IBES, Brown University Role: Postdoctoral Research Associate
2012-2017 (NCE)	Ambient Air Pollution and Incident Stroke (PI: Wellenius) NIH R01-ES020871

Role: Postdoctoral Research Associate

Publications

Articles in Peer-reviewed Journals (*corresponding author)

Year 2020

1. Yang G, **Sun SZ***. Estimation of COVID-19 mortality rate in real time. *Emerging Infectious Diseases* 2020; *Under Revision*.
2. Ran JJ, **Sun SZ**, Han LF, Zhao S, Chen DY, Guo F, Li JH, Qiu H, Lei YJ, Tian LW. Fine particulate matter and cause-specific mortality in the Hong Kong elderly patients with chronic kidney disease. *Chemosphere* 2020; 247: 125913.
3. **Sun SZ**, Cao WN, Chan KP, Ran JJ, Lee RSY, Wong CM, Tian L. Cigarette Smoking Increases Deaths Associated with Air Pollution in Hong Kong. *Atmospheric Environment* 2020; 223: 117266.
4. Cao W[#], **Sun SZ[#]** (**#equal contributors**), Peng L, Gu J, Hao C, Li J, Wei D, Li J. Low willingness to self-pay for pre-exposure prophylaxis (PrEP) among men who have sex with men (MSM) in China. *BMC Public Health* 2020; 20: 337.
5. Ran JJ, Yang AM, **Sun SZ**, Han LF, Qiu H, Mason TG, Li JH, Yang Y, Chan KP, Tian L. Long-term exposure to ambient fine particulate matter and mortality of renal failure: A retrospective cohort study in Hong Kong. *American Journal of Epidemiology* 2020; *In press*.
6. **Sun SZ**, Sarkar C, Kumari S, James P, Cao WN, Lee RSY, Tian LW, Webster C. Air pollution associated respiratory mortality risk alleviated by residential greenness in the Chinese Elderly Health Service Cohort. *Environmental Research* 2020; 183: 109139.
7. Chen D, Mayvaneh F, Baaghdeh M, Entezari A, Ho HC, Xiang Q, Jiao A, Zhang F, Hu K, Chen G, Zhao Q, **Sun SZ**, Ma L, Zhang Y. Utilizing daily excess concentration hours to estimate cardiovascular mortality and years of life lost attributable to fine particulate matter in Tehran, Iran. *Science of the Total Environment* 2020; 703: 134909.

Year 2019

8. Zhang YQ, Xiang QQ, Yu CH, Bao JZ, Ho HC, **Sun SZ**, Ding Z, Hu KJ, Zhang L. Mortality risk and burden associated with temperature variability in China, United Kingdom and United States: Comparative analysis of daily and hourly exposure metrics. *Environmental Research* 2019; 179: 108771.
9. **Sun SZ**, Cao WN, Qiu H, Ran JJ, Lin HL, Shen C, Lee RSY, Tian L. Benefits of physical activity not affected by air pollution: a prospective cohort study. *International Journal of Epidemiology* 2019; doi: 10.1093/ije/dyz184.
10. **Sun SZ**, Cao W, Pun VC, Qiu H, Ge Y, Tian L. Respirable Particulate Constituents and Risks of Cause-specific Mortality in the Hong Kong Population. *Environmental Science & Technology* 2019; 53(16): 9810-9817.
11. **Sun SZ***, Stewart JD, Eliot MN, Yanosky JD, Liao DP, Tinker LF, Eaton CB, Whitsel EA, Wellenius GA (2019). Short-term Exposure to Air Pollution and Incidence of Stroke in the Women's Health Initiative. *Environment International* 2019; 132: 105065.
12. **Sun SZ***, Spangler KR, Weinberger KR, Yanosky JD, Braun JM, Wellenius GA. Ambient Temperature and Markers of Fetal Growth: A Retrospective Observational Study of 29 Million Singleton Births. *Environmental Health Perspectives* 2019; 127(6): 067005. (**Featured in EHP**)
13. Cao WN, Wong HG, Mo PKH, **Sun SZ**, Hu YHJ. A qualitative study of mental health problems among female spouses of HIV-positive men in Sichuan, China. *Journal of the Association of Nurses in AIDS Care* 2019; *In press*.
14. **Sun SZ***, Weinberger KR, Spangler KR, Eliot MN, Braun JM, Wellenius GA. Ambient temperature and preterm birth: A retrospective study of 32 million US singleton births. *Environment International* 2019; 126: 7-13.
15. **Sun SZ**, Tian L, Cao WN, Lai PC, PY Wong, Lee RSY, Mason TG, Tang, R, Kraemer A, Wong CM. Urban Climate Modified Short-term Association of Air pollution with Pneumonia Mortality in Hong

Kong. *Science of The Total Environment* 2019; 646: 618-624.

16. **Sun SZ**, Cao WN, Mason TG, Ran JJ, Qiu H, Li JH, Yang Y, Lin HL, Tian L. Increased susceptibility to heat for respiratory hospitalizations in Hong Kong. *Science of The Total Environment* 2019; 666: 197-204.
17. Lin HL, Guo YF, Ruan ZL, Yang Y, Chen YB, Zheng Y, Cummings-Vaughn LA, Rigdon SE, Vaughn MG, **Sun SZ**, Zhang LL, Wang XJ, Qian ZM, Wu F. Ambient PM_{2.5} and O₃ and their combined effects on prevalence of presbyopia among the elderly: A cross-sectional study in six low- and middle-income countries. *Science of The Total Environment* 2019; 655: 168-173.
18. Tonya GM, Chan KP, Schooling MC, **Sun SZ**, Yang AM, Yang Y, Benjamin B, Tian L. Air quality changes after Hong Kong shipping emission policy: An accountability study, *Chemosphere* 2019; 226: 616-624.

Year 2018

19. **Sun SZ**, Francine L, Jaime H, Qiu H, Wang Y, Wong CM, Lee RSY, Tian L. Seasonal temperature variability and emergency hospital admissions for respiratory diseases: a population-based cohort study. *Thorax* 2018; 73: 951-958.
20. Ran JJ, **Sun SZ**, Yang AM, Yang L, Han LF, Mason TG, Chan KP, Li JH, Tian LW. Effects of ambient benzene and toluene on emergency COPD hospitalizations: a time series study in Hong Kong. *Science of The Total Environment* 2018; 657: 28-35.
21. Dai YR, Qiu H, **Sun SZ**, Yang Y, Lin HL, Tian L. Age-dependent effect of ambient ozone on emergency asthma hospitalizations in Hong Kong. *The Journal of Allergy and Clinical Immunology* 2018; 141(4): 1532-1534.
22. Qiu H, Schooling CM, **Sun SZ**, Tsang H, Tang R, Yang Y, Wong CM, Lee RSY, Tian L. Long-term exposure to fine particulate matter air pollution and Type 2 diabetes mellitus in elderly: a cohort study in Hong Kong. *Environment International* 2018; 113: 350-356.
23. Yang Y, Pun V, **Sun SZ**, Lin HL, Mason TG, Qiu H. Particulate matter components and Health: a literature review on exposure assessment. *Journal of Public Health and Emergency* 2018; 2:14.
24. Ran JJ, Qiu H, **Sun SZ**, Tian L. Short-term effects of ambient TEX (toluene, ethylbenzene, and xylene combined) on cardiorespiratory mortality in Hong Kong. *Environment International* 2018; 117: 91-98.
25. Ran JJ, Qiu H, **Sun SZ**, Yang AM, Tian L. Are ambient volatile organic compounds environmental stressors for heart failure? *Environmental Pollution* 2018; 242: 1810-1816.

Before Year 2018

26. Tian L, Qiu H, **Sun SZ**, Tsang H, Chan KP, Leung WK. Association between Emergency admission for peptic ulcer bleeding and air pollution: A case-crossover analysis in Hong Kong Elderly population. *The Lancet Planetary Health* 2017; 1: e74-81.
27. Qiu H, **Sun SZ**, Tsang H, Wong CM, Lee RSY, Schooling CM, Tian L. Fine particulate matter exposure and incidence of stroke: A cohort study in Hong Kong. *Neurology* 2017; 88: 1-9.
28. Tian L, **Sun SZ**. Comparison of Health Impact of Air Pollution Between China and Other Countries. *Advances in Experimental Medicine and Biology* 2017; 1017:215-232.
29. Tian L, Qiu H, **Sun SZ**, Lin H. Emergency cardiovascular hospitalizations risk attributable to cold temperatures in Hong Kong. *Circulation: Cardiovascular Quality and Outcomes* 2016; 9(2): 135-142.
30. **Sun SZ**, Qiu H, Ho KF, Tian L. Chemical components of respirable particulate matter associated with emergency hospital admissions for Type 2 diabetes in Hong Kong. *Environment International* 2016; 97: 93-99.
31. **Sun SZ**, Tian L, Qiu H, Chan KP, Tsang H, Tang R, Lee RSY, Thach TQ, Wong CM. The influence of pre-existing health conditions on short-term mortality risks of temperature: Evidence from a prospective Chinese elderly cohort in Hong Kong. *Environmental Research* 2016; 148: 7-14.
32. Qiu H[#], **Sun SZ**[#] (**#equal contributors**), Tang R, Chan KP, Tian L. Pneumonia hospitalization risk in the Elderly attributable to cold and hot temperature in Hong Kong. *American Journal of Epidemiology* 2016; 184 (8): 570-578.

33. Li J, **Sun SZ**, Tang R, Qiu H, Huang Q, Mason GM, Tian L. Major air pollutants and COPD: A Systematic review and Meta-analysis. *International Journal of Chronic Obstructive Pulmonary Disease* 2016; 11(1): 3079-3091.
34. **Sun SZ**, Cao P, Chan KP, Tsang H, Wong CM, Thach TQ. Temperature as a modifier of the effects of fine particulate matter on acute mortality in Hong Kong. *Environmental Pollution* 2015; 205: 357-364.

Articles Under Review

35. **Sun SZ**, Weinberger KR, Yan ML, Anderson BG, Wellenius GA. Tropical Cyclone and Risk of Preterm Birth: A Retrospective Analysis of 20 Million Births across 378 US Counties. 2020.
36. **Sun SZ**, Zhao JZ, Cao WN, Zheng TZ, Lu WQ, Zeng Q. Identifying Critical Exposure Windows for Ambient Air Pollution and Semen Quality in Chinese Men. 2020.
37. Yang G, McKay BK, **Sun SZ**, Zhang F, Handel A. Assessing the impact of symptom-based mass screening and testing intervention during a novel infectious disease outbreak: The case of COVID-19. 2020.
38. **Sun SZ**, Cao WN, Ge Yang, Feng S, Zeng Q, Lee RSY, Tian LW, Wellenius GA. Association Between Outdoor Artificial light at Night and Risk of Coronary Heart Disease among Older Adults: A Prospective Cohort Study. 2020.
39. Ran JJ, Kioumourtoglou M, **Sun SZ**, Han LF, Zhao S., Li JH, Chan KP, Tian LW. Source-specific Volatile Organic Compounds and Emergency Hospital Admissions for Cardiorespiratory Diseases. 2020;
40. Zhang M, Liu C, Cui FP, Chen PP, Deng YL, Luo Q, Miao Y, **Sun SZ**, Li YF, Lu WQ, Zeng Q. The Role of Oxidative Stress in Association between Disinfection By-products Exposure and Semen Quality: A Mediation Analysis among Men from an Infertility Clinic. 2020;

Presentations

Conference Presentations

1. Ge Y, Ishikawa Y, Guo YW, **Sun SZ**, and Li CW. The impact of non-acute electronic cigarette uses on cardiovascular system: A systematic review and meta-analysis. 2020 Poster Presentation.
2. **Sun SZ**, Bernstein A, Weinberger KR, Spangler KR, Sheffield P, Wellenius GA. Heat Exposure and Emergency Department Visits to US Children's Hospitals. 2020 Conference of the International Society for Environmental Epidemiology (ISEE) Annual Meeting.
3. **Sun SZ**, Weinberger KR, Yan ML, Anderson BG, Wellenius GA. Tropical Cyclone Exposures and Risk of Preterm Birth in 378 United States Counties: A Nationwide Study. 2020 Conference of the International Society for Environmental Epidemiology (ISEE) Annual Meeting.
4. **Sun SZ**, Weinberger KR, Spangler KR, Eliot MN, Wellenius GA. Ambient temperature and preterm birth: A retrospective observational study of 32 million US singleton births. 2018 ISES-ISEE Joint Annual Meeting, **Oral Presentation**.
5. Mason TG, Chan KP, **Sun SZ**, Qiu H, Yang Y, Barratt B, Tian LW. Air quality changes after Hong Kong's shipping emission policy: An accountability study. 2018 Conference of the International Society for Environmental Epidemiology (ISEE), Poster Presentation.
6. Ran JJ, Qiu H, **Sun SZ**, Yang AM, Tian LW. Are ambient volatile organic compounds environmental stressors for heart failure? 2018 Conference of the International Society for Environmental Epidemiology (ISEE), Poster Presentation.
7. **Sun SZ**, Tian LW, Qiu H, Mason TG, Tang R, Wong CM. PM_{2.5} and mortality: Effect modification by urban microclimate. 2017 Conference of the International Society for Environmental Epidemiology (ISEE), **Oral Presentation**.
8. **Sun SZ**, Qiu H, Crystal CYC, Lai PC, Ruby SYL, Tonya GM, Robert T, Alexander K, Wong CM. Global warming increases mortality effects of air pollution on community-acquired pneumonia. 2016 28th Conference of the International Society for Environmental Epidemiology (ISEE), **Highlighted Poster Presentation**.

9. Yang Y, Tian LW, Qiu H, **Sun SZ**, Chan KP. Ambient nitric oxide associated with increased risk of emergency hospital admission for subarachnoid hemorrhage. 2016 Conference of the International Society for Environmental Epidemiology (ISEE), Poster Presentation.
10. Ran JJ, Qiu H, **Sun SZ**, Tang R, Tian LW. Is air pollution associated with emergency hospitalizations for peptic ulcer diseases? A case-crossover analysis in Hong Kong. 2016 Conference of the International Society for Environmental Epidemiology (ISEE), Poster Presentation.
11. Li JH, **Sun SZ**, Tang R, Qiu H, Tian LW. Main air pollutants and COPD: A systematic review and meta-analysis. 2016 Conference of the International Society for Environmental Epidemiology (ISEE), Poster Presentation.
12. **Sun SZ**, Cao P, Chan KP, Tsang H, Wong CM, Thach TQ. Temperature modifies the effects of fine particulate matter on acute mortality in Hong Kong. 2015 27th Conference of the International Society for Environmental Epidemiology (ISEE), **Oral Presentation**.
13. Wong CM, **Sun SZ**, Tsang H, Chan KP, Lee SY, Thach TQ. Effect modification of long-term exposure on mortality risk associated with short-term exposure to PM_{2.5} in an elderly cohort of Hong Kong. 2015 Conference of the International Society for Environmental Epidemiology (ISEE), Poster Presentation.

Other Presentations

1. Moderate and Extreme Health and Emergency Department Visits to US Children's Hospitals
Boston University, Boston, United States, 2020.
2. Case-crossover Analysis of Air Pollution and Stroke in Women's Health Initiative (WHI)
Women Health Initiative Group Call, 2019.
3. Ambient Temperature and Preterm Birth: A Retrospective Observational Study of 30 Million US Singleton Births. Brown University Center for Environmental Health and Technology (CEHT), 2018.
4. Direct and Indirect Impacts of Temperature on Health in Hong Kong
National Institute of Environmental Health Sciences (NIEHS), 2017.

Service

Editorial Responsibilities

2019 - Present	Associate Editor, <i>BMC Public Health</i>
2019 - Present	Reviewer Board Member, <i>International Journal of Environmental Research and Public Health</i>

Reviewer

American Journal of Respiratory and Critical Care Medicine
Atmospheric Environment
Atmosphere
BMJ
BMJ Open
BMC Public Health
Biomedicine & Pharmacotherapy
Chemosphere
Clinical Infectious Diseases
Environmental Health Perspectives
Environmental Science & Technology
Environment International
European Journal of Epidemiology
Environmental Pollution
Environmental Research
International Journal of Environmental Health Research

International Journal of Environmental Research and Public Health
International Journal of Biometeorology
Journal of Clinical Medicine
Journal of Cleaner Production
JAMA Network Open
Neurology
PLOS Medicine
Paediatric and Perinatal Epidemiology
Science of the Total Environment
The Journal of Infectious Diseases
Tropical Medicine and Infectious Disease

Awards & Fellowships

2017	YS and Christabel Lung Postgraduate Scholarship	The University of Hong Kong
2017	Reaching Out Award	The HKSAR Government
2013-17	Postgraduate Fellowship	The University of Hong Kong
2016	Research Postgraduate Student Exchange Scheme Scholarship	The University of Hong Kong
2016	UGC/CRCG Travel Grant for Research Postgraduate Students	The University of Hong Kong
2016	ISEE Travel Award	ISEE Conference
2015	ISEE Travel Award	ISEE Conference
2015	UGC/CRCG Travel Grant for Research Postgraduate Students	The University of Hong Kong
2008	China National Scholarship	China Ministry of Education

Computer Skills

Advanced skills	R, SAS
Basic skills	ArcGIS, Latex, Python, Java, MySQL